

Convergence of a numerical scheme for solving an inverse problem : Application to tsunami reconstruction

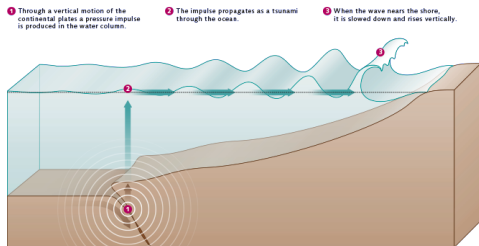
Wave-Ned 2026

Raphaël Terrine (ENSTA-CMAP)

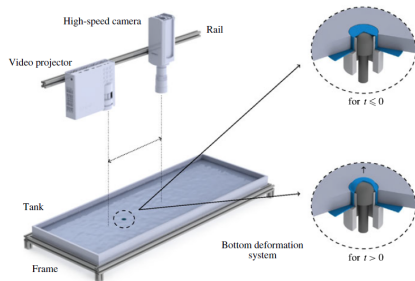
Collaborators : Laurent Bourgeois (UMA) and Philippe Moireau (CMAP)



Motivation



(a) Formation of a tsunami

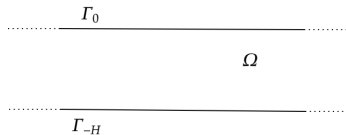


(b) Experimental setup

- Tsunami : seismic wave generated by the displacement of the sea bottom.
- Challenge : quickly detect the sea bottom displacement before disaster.
- Already studied through Tikhonov mixed formulation in the harmonic case, L. Bourgeois, J-F. Mercier, R. Terrine, 2024.
- Problem studied experimentally in a tank, Pablo J. Cobelli et al, 2017.

- 1. Models and studied problem**
- 2. Functional minimisation in an abstract framework**
- 3. Numerical results**
- 4. Conclusions and perspectives**

Complete 2D model and asymptotic limit



g : gravity

c : acoustic celerity

Complete 2D model and asymptotic limit

$$\begin{array}{c}
 \text{.....} \Gamma_0 \text{.....} \\
 \Omega \\
 \text{.....} \Gamma_{-H} \text{.....}
 \end{array}
 \begin{array}{l}
 g : \text{gravity} \\
 c : \text{acoustic celerity}
 \end{array}$$

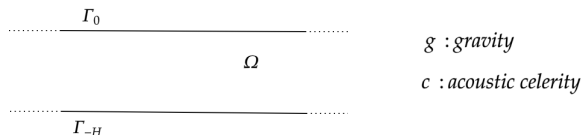
→ Obtained through linearisation of Euler equations^a:

$$\left\{ \begin{array}{ll}
 (\Delta - (1/c^2)\partial_t^2) \phi = 0 & \text{in } \Omega \times (0, T), \\
 \partial_y \phi = \partial_t s & \text{on } \Gamma_0 \times (0, T), \\
 \partial_t \phi + g s = 0 & \text{on } \Gamma_0 \times (0, T), \\
 \partial_\nu \phi = -\partial_t \zeta := \chi & \text{on } \Gamma_{-H} \times (0, T), \\
 \phi(0) = \partial_t \phi(0) = 0 & \text{in } \Omega.
 \end{array} \right.$$

- ϕ : velocity potential.
- s : perturbation of ocean free surface.
- χ : sea-bottom velocity.

^a J. Dubois et al, Well-posedness and potential-based formulation for the propagation of waves, 2024

Complete 2D model and asymptotic limit



→ Obtained through linearisation of Euler equations^a:

$$\begin{cases} (\Delta - (1/c^2)\partial_t^2)\phi = 0 & \text{in } \Omega \times (0, T), \\ \partial_y \phi = \partial_t s & \text{on } \Gamma_0 \times (0, T), \\ \partial_t \phi + g s = 0 & \text{on } \Gamma_0 \times (0, T), \\ \partial_\nu \phi = -\partial_t \zeta := \chi & \text{on } \Gamma_{-H} \times (0, T), \\ \phi(0) = \partial_t \phi(0) = 0 & \text{in } \Omega. \end{cases}$$

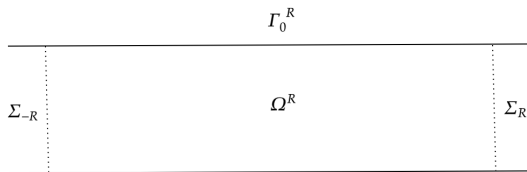
- ϕ : velocity potential.
- s : perturbation of ocean free surface.
- χ : sea-bottom velocity.

Two models :

- Complete model $\implies c < \infty$.
- Gravity model $\implies c = \infty$.

^a J. Dubois et al, Well-posedness and potential-based formulation for the propagation of waves, 2024

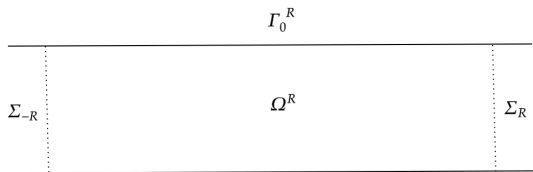
Bounded domain



$$\left\{ \begin{array}{ll} \left(\Delta - (1/c^2) \partial_t^2 \right) \phi = 0 & \text{in } \Omega^R, \\ \partial_y \phi = \partial_t \textcolor{blue}{s} & \text{on } \Gamma_0^R, \\ \partial_t \phi + g \textcolor{blue}{s} = 0 & \text{on } \Gamma_0^R, \\ \partial_\nu \phi = \textcolor{red}{\chi} & \text{on } \Gamma_{-H}^R, \\ \pm \partial_x \phi + \frac{1}{c} \partial_t \phi = 0 & \text{on } \Sigma_{\pm R}^1, \\ \phi(0) = \partial_t \phi(0) = 0 & \text{in } \Omega^R. \end{array} \right.$$

¹K. Dgaygui and P. Joly, *Absorbing boundary conditions for linear gravity waves*, 2006

Bounded domain



$$\left\{ \begin{array}{ll} \left(\Delta - (1/c^2) \partial_t^2 + \nu \Delta \partial_t \right) \phi = 0 & \text{in } \Omega^R, \\ \partial_\nu \phi + \nu \partial_\nu \partial_t \phi = \partial_t s & \text{on } \Gamma_0^R, \\ \partial_t \phi + g s = 0 & \text{on } \Gamma_0^R, \\ \partial_\nu \phi + \nu \partial_\nu \partial_t \phi = \chi & \text{on } \Gamma_{-H}^R, \\ \pm (\partial_x \phi + \nu \partial_\nu \partial_t \phi) + \frac{1}{c} \partial_t \phi = 0 & \text{on } \Sigma_{\pm R}^2, \\ \phi(0) = \partial_t \phi(0) = 0 & \text{in } \Omega^R. \end{array} \right.$$

²K. Dgaygui and P. Joly, *Absorbing boundary conditions for linear gravity waves*, 2006

Abstract framework

Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

Abstract framework

Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

$$W_0(0, T) = \{z \in L^2(0, T; V), \dot{z} \in L^2(0, T; V'), z(0) = 0\}.$$

Abstract framework

Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

$$W_0(0, T) = \{z \in L^2(0, T; V), \dot{z} \in L^2(0, T; V'), z(0) = 0\}.$$

Assumptions :

- For $z, \tilde{z} \in V$, $a_\nu(z, \tilde{z}) = \langle A_\nu z, \tilde{z} \rangle_{V', V}$ and a_ν continuous and λ -coercive.
- Source term $\chi \in L^2(0, T; U)$.
- $B \in \mathcal{L}(U, V')$.

Abstract framework

Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

$$W_0(0, T) = \{z \in L^2(0, T; V), \dot{z} \in L^2(0, T; V'), z(0) = 0\}.$$

Assumptions :

- For $z, \tilde{z} \in V$, $a_\nu(z, \tilde{z}) = \langle A_\nu z, \tilde{z} \rangle_{V', V}$ and a_ν continuous and λ -coercive.
- Source term $\chi \in L^2(0, T; U)$.
- $B \in \mathcal{L}(U, V')$.

Order 1 system : Find $z \in W_0(0, T)$ such that : $\dot{z} + A_\nu z = B\chi$ in $L^2(0, T, V')$.

Abstract framework

Let V, H, U three Hilbert spaces such that $V \hookrightarrow H \equiv H' \hookrightarrow V'$.

$$W_0(0, T) = \{z \in L^2(0, T; V), \dot{z} \in L^2(0, T; V'), z(0) = 0\}.$$

Assumptions :

- For $z, \tilde{z} \in V$, $a_\nu(z, \tilde{z}) = \langle A_\nu z, \tilde{z} \rangle_{V', V}$ and a_ν continuous and λ -coercive.
- Source term $\chi \in L^2(0, T; U)$.
- $B \in \mathcal{L}(U, V')$.

Order 1 system : Find $z \in W_0(0, T)$ such that : $\dot{z} + A_\nu z = B\chi$ in $L^2(0, T, V')$.

Theorem

The problem has a unique solution z^a and:

$$\|z\|_{W_0(0, T)} \leq c \|\chi\|_{L^2(0, T; U)}.$$

^a J-L. Lions E. Magenes, *Non homogeneous boundary value problems and applications*, 1972

Functional framework: order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\phi, \psi, s)^T\|_H^2 = \|\phi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2} \|\psi\|_{L^2(\Omega^R)}^2 + g \|s\|_{L^2(\Gamma_0^R)}^2,$$

Functional framework: order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\phi, \psi, s)^T\|_H^2 = \|\phi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2} \|\psi\|_{L^2(\Omega^R)}^2 + g \|s\|_{L^2(\Gamma_0^R)}^2,$$

- $V = \left\{ (\phi, \psi, s)^T \in H^1(\Omega^R) \times H^1(\Omega^R) \times L^2(\Gamma_0^R) \mid \psi|_{\Gamma_0^R} + gs = 0 \right\}.$

Functional framework: order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\phi, \psi, s)^T\|_H^2 = \|\phi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2} \|\psi\|_{L^2(\Omega^R)}^2 + g \|s\|_{L^2(\Gamma_0^R)}^2,$$

- $V = \left\{ (\phi, \psi, s)^T \in H^1(\Omega^R) \times H^1(\Omega^R) \times L^2(\Gamma_0^R) \mid \psi|_{\Gamma_0^R} + gs = 0 \right\}.$
- $A_\nu \in \mathcal{L}(V, V')$ such that $\forall v, w \in V, \langle A_\nu v, w \rangle_{V', V} = a_\nu(v, w)$ with

$$\begin{aligned} \forall z = (\phi, \psi, s)^T, q = (\mu, \eta, r)^T \in V, \quad a_\nu(z, q) = & -(\psi, \mu)_{H^1(\Omega^R)} + (\nabla \phi, \nabla \eta)_{L^2(\Omega^R)} \\ & + \frac{1}{c} (\psi, \eta)_{L^2(\Sigma_R)} + \frac{1}{c} (\psi, \eta)_{L^2(\Sigma_{-R})} + \nu (\nabla \psi, \nabla \eta)_{L^2(\Omega^R)}. \end{aligned}$$

Functional framework: order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\phi, \psi, s)^T\|_H^2 = \|\phi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2} \|\psi\|_{L^2(\Omega^R)}^2 + g \|s\|_{L^2(\Gamma_0^R)}^2,$$

- $V = \left\{ (\phi, \psi, s)^T \in H^1(\Omega^R) \times H^1(\Omega^R) \times L^2(\Gamma_0^R) \mid \psi|_{\Gamma_0^R} + gs = 0 \right\}.$
- $A_\nu \in \mathcal{L}(V, V')$ such that $\forall v, w \in V, \langle A_\nu v, w \rangle_{V', V} = a_\nu(v, w)$ with

$$\begin{aligned} \forall z = (\phi, \psi, s)^T, q = (\mu, \eta, r)^T \in V, \quad a_\nu(z, q) = & -(\psi, \mu)_{H^1(\Omega^R)} + (\nabla \phi, \nabla \eta)_{L^2(\Omega^R)} \\ & + \frac{1}{c} (\psi, \eta)_{L^2(\Sigma_R)} + \frac{1}{c} (\psi, \eta)_{L^2(\Sigma_{-R})} + \nu (\nabla \psi, \nabla \eta)_{L^2(\Omega^R)}. \end{aligned}$$

- $B \in \mathcal{L}(L^2(\Gamma_{-H}^R), V')$ such that $B^* = \gamma_{\Gamma_{-H}^R}^0.$

Functional framework: order 1 system

- $H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R)$ with the following norm

$$\|(\phi, \psi, s)^T\|_H^2 = \|\phi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2} \|\psi\|_{L^2(\Omega^R)}^2 + g \|s\|_{L^2(\Gamma_0^R)}^2,$$

- $V = \left\{ (\phi, \psi, s)^T \in H^1(\Omega^R) \times H^1(\Omega^R) \times L^2(\Gamma_0^R) \mid \psi|_{\Gamma_0^R} + gs = 0 \right\}.$
- $A_\nu \in \mathcal{L}(V, V')$ such that $\forall v, w \in V, \langle A_\nu v, w \rangle_{V', V} = a_\nu(v, w)$ with

$$\begin{aligned} \forall z = (\phi, \psi, s)^T, q = (\mu, \eta, r)^T \in V, \quad a_\nu(z, q) = & -(\psi, \mu)_{H^1(\Omega^R)} + (\nabla \phi, \nabla \eta)_{L^2(\Omega^R)} \\ & + \frac{1}{c} (\psi, \eta)_{L^2(\Sigma_R)} + \frac{1}{c} (\psi, \eta)_{L^2(\Sigma_{-R})} + \nu (\nabla \psi, \nabla \eta)_{L^2(\Omega^R)}. \end{aligned}$$

- $B \in \mathcal{L}(L^2(\Gamma_{-H}^R), V')$ such that $B^* = \gamma_{\Gamma_{-H}^R}^0.$

Abstract form

The regularised problem solution $z = (\phi, \psi, s)^T \in W_0(0, T)$ is a solution in the variational sense of :

$$\dot{z} + A_\nu z = B\chi \text{ in } L^2(0, T; V').$$

Optimal control approach for solving the inverse problem

Objective

Recovering χ through the measurement y of s in $L^2(0, T, L^2(\Gamma_0^R))$.

Optimal control approach for solving the inverse problem

Objective

Recovering χ through the measurement y of s in $L^2(0, T, L^2(\Gamma_0^R))$.

Tikhonov regularisation : For $\epsilon > 0$:

$$\inf_{\chi \in L^2(0, T; L^2(\Gamma_{-H}^R))} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_{L^2(\Gamma_0^R)}^2 + \frac{\epsilon}{2} \|\chi(t)\|_{L^2(\Gamma_{-H}^R)}^2 dt,$$

with $\forall z = (\phi, \psi, s)^T \in H$, $Cz = s$ and

$$\begin{cases} \dot{z}|_{\chi} + A_{\nu} z|_{\chi} = B\chi, \\ z|_{\chi}(0) = 0, \end{cases}$$

Optimal control approach for solving the inverse problem : abstract framework

Let Y, V, H, U four Hilbert spaces with $V \subset H \subset V'$. Let $C \in \mathcal{L}(H, Y)$: measurement operator.

Objective

Recover χ through the measurement y of Cz .

Optimal control approach for solving the inverse problem : abstract framework

Let Y, V, H, U four Hilbert spaces with $V \subset H \subset V'$. Let $C \in \mathcal{L}(H, Y)$: measurement operator.

Objective

Recover χ through the measurement y of Cz .

Optimal control problem : For $\epsilon > 0$:

$$\inf_{\chi \in L^2(0, T; U)} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_Y^2 + \frac{\epsilon}{2} \|\chi(t)\|_U^2 dt,$$

with $z|_{\chi}$ the solution in $W(0, T)$ of $\dot{z}|_{\chi} + A_{\nu} z|_{\chi} = B\chi$ and $z|_{\chi}(0) = 0$.

Optimal control approach for solving the inverse problem : abstract framework

Let Y, V, H, U four Hilbert spaces with $V \subset H \subset V'$. Let $C \in \mathcal{L}(H, Y)$: measurement operator.

Objective

Recover χ through the measurement y of Cz .

Optimal control problem : For $\epsilon > 0$:

$$\inf_{\chi \in L^2(0, T; U)} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_Y^2 + \frac{\epsilon}{2} \|\chi(t)\|_U^2 dt,$$

with $z|_{\chi}$ the solution in $W(0, T)$ of $\dot{z}|_{\chi} + A_{\nu} z|_{\chi} = B\chi$ and $z|_{\chi}(0) = 0$.

Well-posedness

The previous problem has a unique solution : $\mathcal{J}$ continuous, coercive, strictly convex.

Adjoint state of discretised criterion

$$\mathcal{J}_\tau^h(\{\chi_h^n\}_{1 \leq n \leq N}) = \sum_{n=0}^N \frac{\tau}{2} \|y_h^n - Cz_h^n\|_Y^2 + \sum_{n=1}^N \frac{\tau \epsilon}{2} \|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n - B\chi_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket \\ z_h^0 = 0. \end{cases}$$

Adjoint state of discretised criterion

$$\mathcal{J}_\tau^h(\{\chi_h^n\}_{1 \leq n \leq N}) = \sum_{n=0}^N \frac{\tau}{2} \|y_h^n - Cz_h^n\|_Y^2 + \sum_{n=1}^N \frac{\tau \epsilon}{2} \|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n - B\chi_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket \\ z_h^0 = 0. \end{cases}$$

$$\begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n = B\chi_h^n, & \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* p_h^n = C^* (y_h^n - Cz_h^n), & \forall n \in \llbracket 1; N \rrbracket, \\ \epsilon \chi_h^n - B^* p_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket, \\ z_h^0 = p_h^{N+1} = 0. \end{cases}$$

Adjoint state of discretised criterion

$$\mathcal{J}_\tau^h(\{\chi_h^n\}_{1 \leq n \leq N}) = \sum_{n=0}^N \frac{\tau}{2} \|y_h^n - Cz_h^n\|_Y^2 + \sum_{n=1}^N \frac{\tau \epsilon}{2} \|\chi_h^n\|_U^2,$$

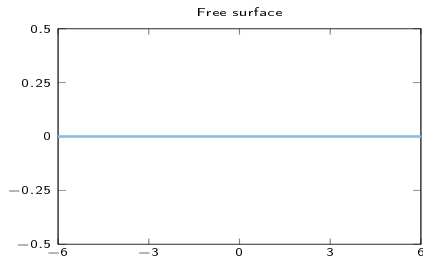
$$\text{with } \begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n - B\chi_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket \\ z_h^0 = 0. \end{cases}$$

$$\begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n = B\chi_h^n, & \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* p_h^n = C^* (y_h^n - Cz_h^n), & \forall n \in \llbracket 1; N \rrbracket, \\ \epsilon \chi_h^n - B^* p_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket, \\ z_h^0 = p_h^{N+1} = 0. \end{cases}$$

Solving the problem as an engineer

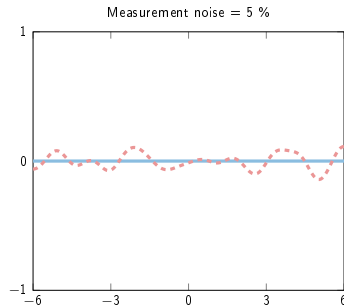
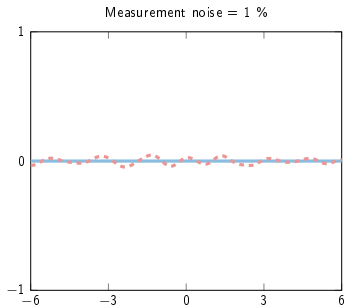
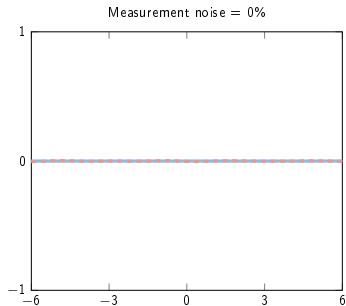
- Compute analytically the gradient of $\mathcal{J}_\tau^h$.
- Use a gradient descent algorithm to find the minimiser.
- We use L-BFGS for fast convergence.

Numerical reconstruction



Parameters

- $R = 90cm$.
- $H = 15cm$.
- $h = 0.1, \mathbb{P}_2$ elements.
- $\tau = h^2$.
- $\epsilon = \nu = 10^{-6}$.



Our contribution : Numerical analysis of the scheme

$$\inf_{\chi \in L^2(0,T;U)} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_Y^2 + \frac{\epsilon}{2} \|\chi(t)\|_U^2 dt,$$

with $z|_{\chi}$ the solution in $W(0, T)$ of $\dot{z}|_{\chi} + A_{\nu} z|_{\chi} = B\chi$ and $z|_{\chi}(0) = 0$.

Our contribution : Numerical analysis of the scheme

$$\inf_{\chi \in L^2(0,T;U)} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_Y^2 + \frac{\epsilon}{2} \|\chi(t)\|_U^2 dt,$$

with $z|_{\chi}$ the solution in $W(0,T)$ of $\dot{z}|_{\chi} + A_{\nu}z|_{\chi} = B\chi$ and $z|_{\chi}(0) = 0$.

$$\mathcal{J}_{\tau}^h(\{\chi_h^n\}_{1 \leq n \leq N}) = \sum_{n=0}^N \frac{\tau}{2} \|y_h^n - Cz_h^n\|_Y^2 + \sum_{n=1}^N \frac{\tau\epsilon}{2} \|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_{\nu}z_h^n - B\chi_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket \\ z_h^0 = 0. \end{cases}$$

Our contribution : Numerical analysis of the scheme

$$\inf_{\chi \in L^2(0,T;U)} \mathcal{J}(\chi) = \int_0^T \frac{1}{2} \|y(t) - Cz|_{\chi}(t)\|_Y^2 + \frac{\epsilon}{2} \|\chi(t)\|_U^2 dt,$$

with $z|_{\chi}$ the solution in $W(0, T)$ of $\dot{z}|_{\chi} + A_{\nu} z|_{\chi} = B\chi$ and $z|_{\chi}(0) = 0$.

$$\mathcal{J}_{\tau}^h(\{\chi_h^n\}_{1 \leq n \leq N}) = \sum_{n=0}^N \frac{\tau}{2} \|y_h^n - Cz_h^n\|_Y^2 + \sum_{n=1}^N \frac{\tau \epsilon}{2} \|\chi_h^n\|_U^2,$$

$$\text{with } \begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_{\nu} z_h^n - B\chi_h^n = 0, & \forall n \in \llbracket 1; N \rrbracket \\ z_h^0 = 0. \end{cases}$$

Questions

- $\{\chi_h^n\}_{1 \leq n \leq N} \rightarrow \bar{\chi}$ when $(\tau, h) \rightarrow 0$?
- At which rate ?

Space-time mixed formulation

Find $u = (z, \chi)^T \in W_0(0, T) \times L^2(0, T, U)$ and $p \in L^2(0, T; V)$ such that for all $v = (\tilde{z}, \eta)^T \in W_0(0, T) \times L^2(0, T, U)$ and for all $q \in L^2(0, T; V)$:

$$\begin{cases} j(u, v) + d(v, p) = \ell(v), \\ d(u, q) = 0, \end{cases}$$

$$j(u, v) = \int_0^T \epsilon(\chi, \eta)_U + (Cz, C\tilde{z})_Y \, dt, \quad d(u, q) = \int_0^T \langle \dot{z}, q \rangle_{V', V} + a_\nu(z, q) - b(\chi, q) \, dt$$

$$\ell(v) = \int_0^T (y, C\tilde{z})_Y \, dt.$$

Well-posedness of the mixed formulation

Coercivity of j on $\ker d$

There exists $C > 0$ such that for all $u \in W_0(0, T) \times L^2(0, T; U)$ such that $d(u, q) = 0, \forall q \in L^2(0, T; V)$, we get :

$$j(u, u) \geq C \|u\|_{W_0(0, T) \times L^2(0, T; U)}^2$$

Well-posedness of the mixed formulation

Coercivity of j on $\ker d$

There exists $C > 0$ such that for all $u \in W_0(0, T) \times L^2(0, T; U)$ such that $d(u, q) = 0, \forall q \in L^2(0, T; V)$, we get :

$$j(u, u) \geq C \|u\|_{W_0(0, T) \times L^2(0, T; U)}^2$$

Inf-sup condition

There exists $\beta > 0$ such that:

$$\inf_{p \in L^2(0, T; V)} \sup_{v \in W_0(0, T) \times L^2(0, T; U)} \frac{d(v, p)}{\|v\|_{W_0(0, T) \times L^2(0, T; U)} \|p\|_{L^2(0, T; V)}} \geq \beta.$$

Well-posedness of the mixed formulation

Coercivity of j on $\ker d$

There exists $C > 0$ such that for all $u \in W_0(0, T) \times L^2(0, T; U)$ such that $d(u, q) = 0, \forall q \in L^2(0, T; V)$, we get :

$$j(u, u) \geq C \|u\|_{W_0(0, T) \times L^2(0, T; U)}^2$$

Inf-sup condition

There exists $\beta > 0$ such that:

$$\inf_{p \in L^2(0, T; V)} \sup_{v \in W_0(0, T) \times L^2(0, T; U)} \frac{d(v, p)}{\|v\|_{W_0(0, T) \times L^2(0, T; U)} \|p\|_{L^2(0, T; V)}} \geq \beta.$$

Fundamental result : Equivalence between problems

χ the solution of the mixed formulation is the solution $\bar{\chi}$ of the optimal control problem.

Finite elements discretisation

- **Discretisation spaces** : $V_h \subset V$, $U_h \subset U$ et $Y_h \subset Y$ for $h > 0$.
- $W_h^\tau = \{w_h^\tau \in \mathbb{P}_1 \otimes V_h, w_h^\tau(0) = 0\}$.

Finite elements discretisation

- **Discretisation spaces** : $V_h \subset V$, $U_h \subset U$ et $Y_h \subset Y$ for $h > 0$.
- $W_h^\tau = \{w_h^\tau \in \mathbb{P}_1 \otimes V_h, w_h^\tau(0) = 0\}$.

Discrete mixed formulation

Find $u_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and $p_h^\tau \in \mathbb{P}_0 \otimes V_h$ such that for all $v_h^\tau = (\tilde{z}_h^\tau, \eta_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and for all $q_h^\tau \in \mathbb{P}_0 \otimes V_h$:

$$\begin{cases} j(u_h^\tau, v_h^\tau) + d(v_h^\tau, p_h^\tau) = \ell(v_h^\tau), \\ d(u_h^\tau, q_h^\tau) = 0, \end{cases}$$

Finite elements discretisation

- **Discretisation spaces** : $V_h \subset V$, $U_h \subset U$ et $Y_h \subset Y$ for $h > 0$.
- $W_h^\tau = \{w_h^\tau \in \mathbb{P}_1 \otimes V_h, w_h^\tau(0) = 0\}$.

Discrete mixed formulation

Find $u_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and $p_h^\tau \in \mathbb{P}_0 \otimes V_h$ such that for all $v_h^\tau = (\tilde{z}_h^\tau, \eta_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and for all $q_h^\tau \in \mathbb{P}_0 \otimes V_h$:

$$\begin{cases} j(u_h^\tau, v_h^\tau) + d(v_h^\tau, p_h^\tau) = \ell(v_h^\tau), \\ d(u_h^\tau, q_h^\tau) = 0, \end{cases}$$

$$z_h^\tau(x, t) = \sum_{n=1}^N z_h^n(x) \omega^n(t); \chi_h^\tau(x, t) = \sum_{n=1}^N \chi_h^n(x) \omega_0^n(t); y_h^\tau(x, t) = \sum_{n=1}^N y_h^n(x) \omega_0^n(t), p_h^\tau(x, t) = \sum_{n=1}^N p_h^n(x) \omega_0^n(t).$$

Conforming time scheme

Time scheme with conforming FEM

$$\left\{ \begin{array}{l} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu \frac{z_h^n + z_h^{n-1}}{2} = B\chi_h^n, \quad \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* \left(\frac{p_h^{n+1} + p_h^n}{2} \right) = C^* \left(\frac{y_h^{n-1} + y_h^n}{2} - C \frac{z_h^{n-1} + 4z_h^n + z_h^{n+1}}{6} \right), \quad \forall n \in \llbracket 1; N-1 \rrbracket \\ \frac{p_h^N - p_h^{N+1}}{\tau} + A_\nu^* \left(\frac{p_h^{N+1} + p_h^N}{2} \right) = C^* \left(\frac{y_h^{N-1} + y_h^N}{2} - C \left(\frac{1}{6} z_h^{N-1} + \frac{1}{3} z_h^N \right) \right), \\ \epsilon \chi_h^n - B^* p_h^n = 0, \quad \forall n \in \llbracket 1; N \rrbracket, \\ p_h^{N+1} = z_h^0 = 0. \end{array} \right.$$

Conforming time scheme

Time scheme with conforming FEM

$$\left\{ \begin{array}{l} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu \frac{z_h^n + z_h^{n-1}}{2} = B\chi_h^n, \quad \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* \left(\frac{p_h^{n+1} + p_h^n}{2} \right) = C^* \left(\frac{y_h^{n-1} + y_h^n}{2} - C \frac{z_h^{n-1} + 4z_h^n + z_h^{n+1}}{6} \right), \quad \forall n \in \llbracket 1; N-1 \rrbracket \\ \frac{p_h^N - p_h^{N+1}}{\tau} + A_\nu^* \left(\frac{p_h^{N+1} + p_h^N}{2} \right) = C^* \left(\frac{y_h^{N-1} + y_h^N}{2} - C \left(\frac{1}{6} z_h^{N-1} + \frac{1}{3} z_h^N \right) \right), \\ \epsilon \chi_h^n - B^* p_h^n = 0, \quad \forall n \in \llbracket 1; N \rrbracket, \\ p_h^{N+1} = z_h^0 = 0. \end{array} \right.$$

- Data attachment term different from that of the Euler implicit scheme.
- Crank-Nicholson scheme rather than implicit Euler scheme (no uniform energy identity).

Adding a quadrature rule

With $\oint_0^T f(t) dt = \tau \sum_{n=1}^N f(n\tau_-)$:

- $j(u_h^\tau, v_h^\tau) \rightarrow j_\tau(u_h^\tau, v_h^\tau) = \int_0^T \epsilon(\partial_t \chi_h^\tau, \partial_t \eta_h^\tau)_U dt + \oint_0^T (Cz_h^\tau, C\tilde{z}_h^\tau)_Y dt.$
- $\ell(v_h^\tau) \rightarrow \ell_\tau(v_h^\tau) = \oint_0^T (y, C\tilde{z}_h^\tau)_Y dt.$
- $d(u_h^\tau, q_h^\tau) \rightarrow d_\tau(u_h^\tau, q_h^\tau) = \int_0^T \langle \dot{z}_h^\tau, q_h^\tau \rangle_{V',V} - b(\chi_h^\tau, q_h^\tau) dt + \oint_0^T +a_\nu(z_h^\tau, q_h^\tau) dt.$

Adding a quadrature rule

With $\oint_0^T f(t) dt = \tau \sum_{n=1}^N f(n\tau_-)$:

- $j(u_h^\tau, v_h^\tau) \rightarrow j_\tau(u_h^\tau, v_h^\tau) = \int_0^T \epsilon(\partial_t \chi_h^\tau, \partial_t \eta_h^\tau)_U dt + \oint_0^T (Cz_h^\tau, C\tilde{z}_h^\tau)_Y dt.$
- $\ell(v_h^\tau) \rightarrow \ell_\tau(v_h^\tau) = \oint_0^T (y, C\tilde{z}_h^\tau)_Y dt.$
- $d(u_h^\tau, q_h^\tau) \rightarrow d_\tau(u_h^\tau, q_h^\tau) = \int_0^T \langle \dot{z}_h^\tau, q_h^\tau \rangle_{V',V} - b(\chi_h^\tau, q_h^\tau) dt + \oint_0^T +a_\nu(z_h^\tau, q_h^\tau) dt.$

Time scheme with quadrature rule

$$\begin{cases} \frac{z_h^n - z_h^{n-1}}{\tau} + A_\nu z_h^n = B\chi_h^n, & \forall n \in \llbracket 1; N \rrbracket, \\ \frac{p_h^n - p_h^{n+1}}{\tau} + A_\nu^* p_h^n = C^* (y_h^n - Cz_h^n), & \forall n \in \llbracket 1; N \rrbracket \\ \epsilon \chi_h^n - B^* p_h^n = 0 & \forall n \in \llbracket 1; N \rrbracket, \\ p_h^{N+1} = z_h^0 = 0. \end{cases}$$

Technical results

Assumption : quasi-uniform mesh

We suppose that the mesh is "shape regular" and there exists $c > 0$ such that:

$$\forall h > 0, \forall K \in \mathcal{T}_h, ch \leq h_K := \text{Diam}(K) \leq h.$$

Technical results

Assumption : quasi-uniform mesh

We suppose that the mesh is "shape regular" and there exists $c > 0$ such that:

$$\forall h > 0, \forall K \in \mathcal{T}_h, ch \leq h_K := \text{Diam}(K) \leq h.$$

For all $v_h \in V_h$, we introduce the following norm:

$$\|v_h\|_{V'_h} := \sup_{w_h \in V_h} \frac{|\langle v_h, w_h \rangle_{V', V}|}{\|w_h\|_V}.$$

Technical results

Assumption : quasi-uniform mesh

We suppose that the mesh is "shape regular" and there exists $c > 0$ such that:

$$\forall h > 0, \forall K \in \mathcal{T}_h, ch \leq h_K := \text{Diam}(K) \leq h.$$

For all $v_h \in V_h$, we introduce the following norm:

$$\|v_h\|_{V'_h} := \sup_{w_h \in V_h} \frac{|\langle v_h, w_h \rangle_{V', V}|}{\|w_h\|_V}.$$

Equivalence of norms

For all $v_h \in V_h$, we have $\|v_h\|_{V'_h} \leq \|v_h\|_{V'} \leq \|P_h\|_{\mathcal{L}(V)} \|v_h\|_{V'_h}$ with

$$P_h : V \rightarrow V_h, P_h(z) = \arg \min_{z_h \in V_h} \|z - z_h\|_H.^a$$

^aA.Ern J-L.Guermond, *Finite Elements III*, 2021

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

Find $u_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and $p_h^\tau \in \mathbb{P}_0 \otimes V_h$ such that for all $v_h^\tau = (\tilde{z}_h^\tau, \eta_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and for all $q_h^\tau \in \mathbb{P}_0 \otimes V_h$:

$$\begin{cases} j_\tau(u_h^\tau, v_h^\tau) + d_\tau(v_h^\tau, p_h^\tau) = \ell_\tau(v_h^\tau), \\ d_\tau(u_h^\tau, q_h^\tau) = 0. \end{cases}$$

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

Find $u_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and $p_h^\tau \in \mathbb{P}_0 \otimes V_h$ such that for all $v_h^\tau = (\tilde{z}_h^\tau, \eta_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and for all $q_h^\tau \in \mathbb{P}_0 \otimes V_h$:

$$\begin{cases} j_\tau(u_h^\tau, v_h^\tau) + d_\tau(v_h^\tau, p_h^\tau) = \ell_\tau(v_h^\tau), \\ d_\tau(u_h^\tau, q_h^\tau) = 0. \end{cases}$$

- j_τ coercive on $\ker d_\tau$ with a constant α that depends on $\|P_h\|_{\mathcal{L}(V)}$.
- Inf-sup condition for d_τ with constant β that depends on $\|P_h\|_{\mathcal{L}(V)}$.

Well-posedness of the discrete problem

Discrete mixed formulation with quadrature

Find $u_h^\tau = (z_h^\tau, \chi_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and $p_h^\tau \in \mathbb{P}_0 \otimes V_h$ such that for all $v_h^\tau = (\tilde{z}_h^\tau, \eta_h^\tau)^T \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)$ and for all $q_h^\tau \in \mathbb{P}_0 \otimes V_h$:

$$\begin{cases} j_\tau(u_h^\tau, v_h^\tau) + d_\tau(v_h^\tau, p_h^\tau) = \ell_\tau(v_h^\tau), \\ d_\tau(u_h^\tau, q_h^\tau) = 0. \end{cases}$$

- j_τ coercive on $\ker d_\tau$ with a constant α that depends on $\|P_h\|_{\mathcal{L}(V)}$.
- Inf-sup condition for d_τ with constant β that depends on $\|P_h\|_{\mathcal{L}(V)}$.

Well-posedness

The discrete mixed formulation with quadrature is well-posed.^a

^aF. Brezzi, M. Fortin, *Mixed and hybrid finite element methods*, 1991

Convergence rate

Second Strang Lemma

$$\begin{aligned} \|u_h^\tau - u\|_{W_0(0,T) \times L^2(0,T;U)} &\leq C \|P_h\|_{\mathcal{L}(V)}^3 \left(\inf_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \|u - v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)} \right. \\ &+ \inf_{q_h^\tau \in \mathbb{P}_0 \otimes V_h} \|p - q_h^\tau\|_{L^\infty(0,T;V)} + \inf_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \|u - v_h^\tau\|_{L^\infty(0,T;V) \times L^2(0,T;U)} \\ &+ \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|j(u, v_h^\tau) - j_\tau(u, v_h^\tau)|}{\|v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)}} + \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|d(v_h^\tau, p) - d_\tau(v_h^\tau, p)|}{\|v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)}} \\ &+ \left. \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|\ell(v_h^\tau) - \ell_\tau(v_h^\tau)|}{\|v_h^\tau\|_{W_0(0,T) \times H_1^1(0,T;U)}} + \sup_{q_h^\tau \in \mathbb{P}_0 \otimes V_h} \frac{|d(u, q_h^\tau) - d_\tau(u, q_h^\tau)|}{\|q_h^\tau\|_{L^2(0,T;V)}} \right) \end{aligned}$$

Convergence rate

Second Strang Lemma

$$\begin{aligned} \|u_h^\tau - u\|_{W_0(0,T) \times L^2(0,T;U)} &\leq C \|P_h\|_{\mathcal{L}(V)}^3 \left(\inf_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \|u - v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)} \right. \\ &+ \inf_{q_h^\tau \in \mathbb{P}_0 \otimes V_h} \|p - q_h^\tau\|_{L^\infty(0,T;V)} + \inf_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \|u - v_h^\tau\|_{L^\infty(0,T;V) \times L^2(0,T;U)} \\ &+ \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|j(u, v_h^\tau) - j_\tau(u, v_h^\tau)|}{\|v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)}} + \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|d(v_h^\tau, p) - d_\tau(v_h^\tau, p)|}{\|v_h^\tau\|_{W_0(0,T) \times L^2(0,T;U)}} \\ &+ \sup_{v_h^\tau \in W_h^\tau \times (\mathbb{P}_0 \otimes U_h)} \frac{|\ell(v_h^\tau) - \ell_\tau(v_h^\tau)|}{\|v_h^\tau\|_{W_0(0,T) \times H_1^1(0,T;U)}} + \sup_{q_h^\tau \in \mathbb{P}_0 \otimes V_h} \frac{|d(u, q_h^\tau) - d_\tau(u, q_h^\tau)|}{\|q_h^\tau\|_{L^2(0,T;V)}} \Big) \end{aligned}$$

Convergence error for $\mathbb{P}_k$ Lagrange finite elements

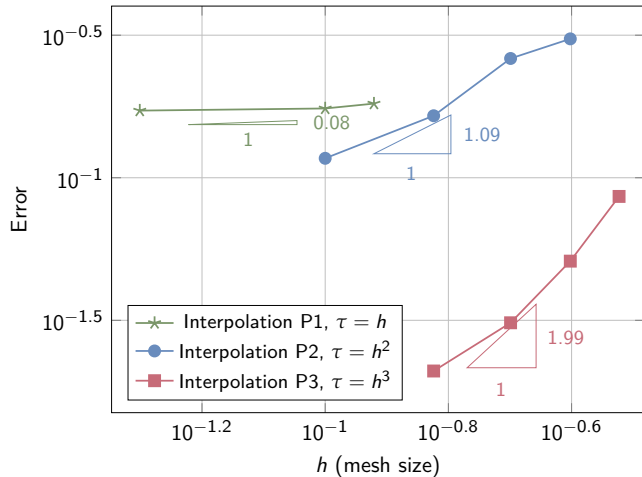
$$\sqrt{\|z_h^\tau - z\|_{W_0(0,T)}^2 + \|\chi_h^\tau - \chi\|_{L^2(0,T;U)}^2} \leq C \|P_h\|_{\mathcal{L}(V)}^3 (\tau + h^k)$$

Observed rate of convergence

$$\|\chi_h^\tau - \chi\|_{L^2(0,\tau;U)} \leq \frac{C}{h^{\frac{3}{2}}}(\tau + h^k)$$

Observed rate of convergence

$$\|\chi_h^\tau - \chi\|_{L^2(0,T;U)} \leq \frac{C}{h^{\frac{3}{2}}}(\tau + h^k)$$



Perspectives without dissipation

- Joint work with J. Dubois and S. Impériale

Perspectives without dissipation

- Joint work with J. Dubois and S. Impériale
- We rewrite the initial model as a Friedrich's system :

$$\begin{cases} \partial_t U - G^* V = 0 \\ \partial_t V + GU = B \int_0^t \chi(s) ds \text{ where } B : H^{-\frac{1}{2}}(\Gamma_{-H+\theta}) \rightarrow \mathcal{G} \text{ s.t. } (B\chi, (u, v))_{\mathcal{G}} = \langle \chi, u \rangle \\ U(0) = V(0) = 0 \end{cases}$$

Perspectives without dissipation

- Joint work with J. Dubois and S. Impériale
- We rewrite the initial model as a Friedrich's system :

$$\begin{cases} \partial_t U - G^* V = 0 \\ \partial_t V + GU = B \int_0^t \chi(s) ds \text{ where } B : H^{-\frac{1}{2}}(\Gamma_{-H+\theta}) \rightarrow \mathcal{G} \text{ s.t. } (B\chi, (u, v))_{\mathcal{G}} = \langle \chi, u \rangle \\ U(0) = V(0) = 0 \end{cases}$$

- Where we have the inf-sup condition at the continuous level

Perspectives without dissipation

- Joint work with J. Dubois and S. Impériale
- We rewrite the initial model as a Friedrich's system :

$$\begin{cases} \partial_t U - G^* V = 0 \\ \partial_t V + GU = B \int_0^t \chi(s) ds \text{ where } B : H^{-\frac{1}{2}}(\Gamma_{-H+\theta}) \rightarrow \mathcal{G} \text{ s.t. } (B\chi, (u, v))_{\mathcal{G}} = \langle \chi, u \rangle \\ U(0) = V(0) = 0 \end{cases}$$

- Where we have the [inf-sup condition at the continuous level](#)
- Then the optimality system associated with the least squares minimization can be reformulated as a mixed formulation

Perspectives without dissipation

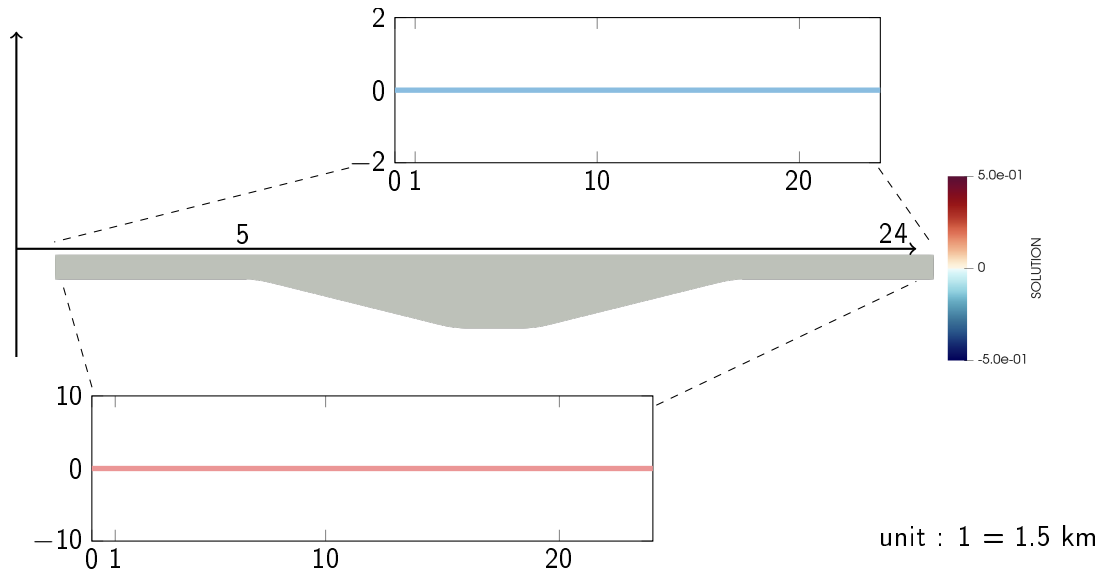
- Joint work with J. Dubois and S. Impériale
- We rewrite the initial model as a Friedrich's system :

$$\begin{cases} \partial_t U - G^* V = 0 \\ \partial_t V + GU = B \int_0^t \chi(s) ds \text{ where } B : H^{-\frac{1}{2}}(\Gamma_{-H+\theta}) \rightarrow \mathcal{G} \text{ s.t. } (B\chi, (u, v))_{\mathcal{G}} = \langle \chi, u \rangle \\ U(0) = V(0) = 0 \end{cases}$$

- Where we have the inf-sup condition at the continuous level
- Then the optimality system associated with the least squares minimization can be reformulated as a mixed formulation

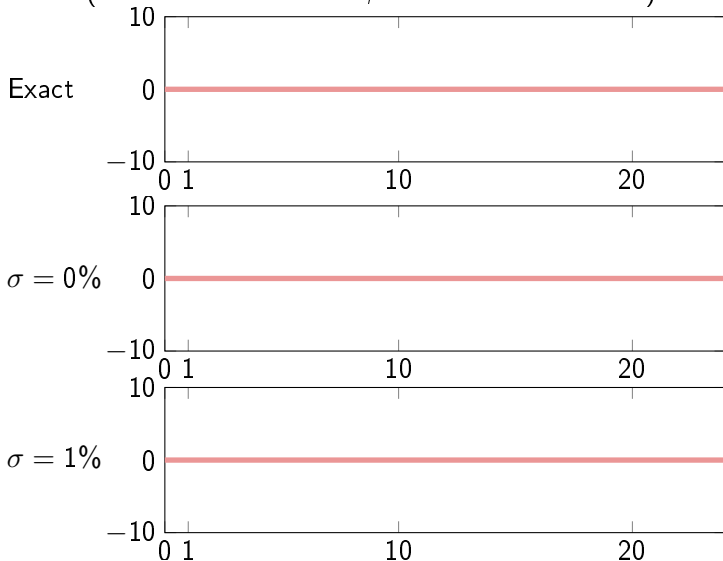
However, efficient discretization with high-order spatial finite elements and an explicit scheme will not satisfy a discrete inf-sup condition.

Direct problem



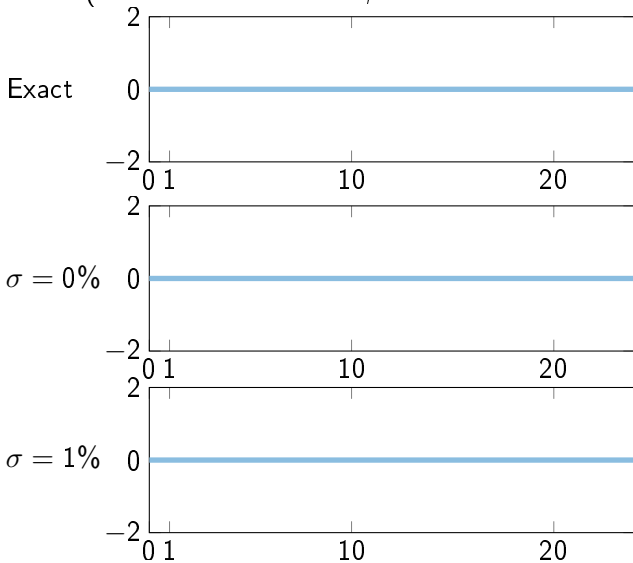
Reconstruction of the bottom source

- Reconstruction: (321 dof for the source, 4000 BFGS iterations) – Slow-down visualization



Associated free surface

- Reconstruction: (321 dof for the source, 4000 BFGS iterations) – Slow-down visualization



Thank you !



Website : <https://raphaelterriner.github.io/>

Mail : raphael.terriner@ensta.fr

Choice of ϵ with respect to the noise level

